Lab Report

X-ray Powder Diffraction of Silicon

NAINI DIWAN

Roll no.: 23199

Abstract

Powder X-ray diffraction patterns of silicon were recorded with Cu Kα radiation (3°–90° in 0.01° steps) in two separate runs, in order to index the pattern and determine the cubic lattice parameter. Six reflections were observed in each run. The Kα_1/Kα_2 doublets were fitted with pseudo-Voigt profiles, giving Kα_1 peak positions to 0.0002–0.003°. With the mathematical and analytical indexing methods, the ratios $\sin^2\theta_i/\sin^2\theta_1 = 3:8:11:16:19:24$ identify a face-centred cubic lattice, with (111), (220), (311), (400), (331) and (422) present and (200), (222), (420) absent, as expected for the diamond structure. A least-squares fit of all peaks with a free zero shift gives $a = 5.43220 \pm 0.00016$ Å (Run 1) and 5.4304 ± 0.0014 Å (Run 2), against the literature value 5.43093 ± 0.00002 Å. Run 2 agrees with the literature value within 0.4σ; Run 1 lies 0.023% above it, more than its statistical uncertainty allows, which points to an uncorrected systematic error of about 1×10^{-3} Å. Both runs reproduce the lattice parameter of silicon to within 0.05%.

Contents

1 Introduction **3**

2 Theory **3**

3 Experimental Setup and Procedure **5**
3.1 Apparatus 5
3.2 Procedure 5
3.3 Sources of Uncertainty 5

4 Observations and Data **6**

5 Analysis and Results **7**
5.1 Data Reduction 7
5.2 Fitting and Error Propagation 7
5.3 Result Summary 11

6 Discussion **11**

7 Conclusion **12**

A Raw Data and Worked Calculations **12**
A.1 Raw data and fit parameters 12
A.2 Calculations of a and σ_a 13
A.3 Per-peak results 14
A.4 Quotient search (analytical method) 15
A.5 Least-squares calculations 16

1 Introduction

Crystalline solids diffract X-rays because their atoms form a periodic array whose plane spacings are comparable to the X-ray wavelength. Bragg's law [1] relates the diffraction angles to the plane spacings, so that the positions of the lines in a powder pattern fix the size and shape of the unit cell, while their intensities and systematic absences reflect the arrangement of atoms inside it [2]. Silicon is a convenient test case: it crystallizes in the diamond-cubic structure (space group $Fd\bar{3}m$) with $a \approx 5.431$ Å [4], a high-symmetry structure that can be indexed by hand.

The patterns analysed in this report were recorded on silicon powder (the peak positions of both runs match silicon, Sec. 4). The aims of the experiment were:

- To measure the X-ray powder diffraction pattern of Si as a function of the scattering angle 2θ and extract the Kα_1 peak positions.
- To index the peaks by the mathematical and the analytical method, and to identify the Bravais lattice from the sequence of $h^2 + k^2 + l^2$.
- To determine the cubic lattice parameter a with its uncertainty.

2 Theory

Constructive interference from a family of lattice planes with spacing d_{hkl} occurs when

$$2\, d_{hkl} \sin\theta = \lambda, \tag{1}$$

where θ is the Bragg angle (half the scattering angle 2θ), λ is the X-ray wavelength (Å), and the diffraction order is absorbed into the Miller indices hkl.

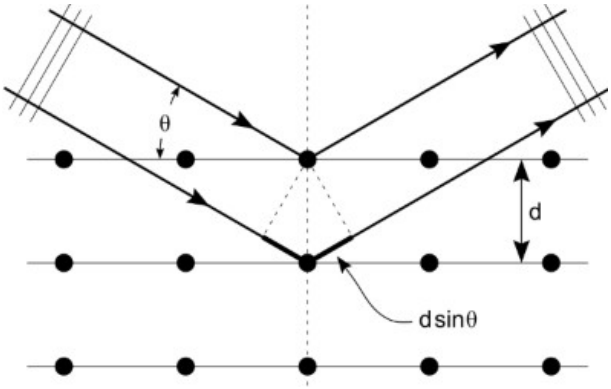


Figure 1: **Bragg diffraction:** Two beams with identical wavelength and phase approach a crystalline solid and are scattered off two different atoms within it.

For a cubic lattice with lattice parameter a (Å),

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}. \tag{2}$$

Combining Eqs. (1) and (2) gives

$$\sin^2\theta = \frac{\lambda^2}{4a^2}\,(h^2 + k^2 + l^2) = K\,N, \qquad K \equiv \frac{\lambda^2}{4a^2}, \quad N \equiv h^2 + k^2 + l^2. \tag{3}$$

Here N is a positive integer and K is the same constant for every line of a given pattern.

Mathematical method. The ratios

$$\frac{\sin^2\theta_i}{\sin^2\theta_1} = \frac{N_i}{N_1} \tag{4}$$

are multiplied by the smallest integer $(1, 2, 3, \dots)$ that makes all of them integers; this gives N_i/N_1 times that integer, and the integers are then identified with N_i using the allowed sequences of Table 1.

Analytical method. Quotients $\sin^2\theta_i/n$ ($n = 1, 2, 3, \dots$) are tabulated. The lowest value K that appears in the list of every peak is the constant of Eq. (3), and $N_i = \sin^2\theta_i/K$.

Table 1: Allowed values of $N = h^2 + k^2 + l^2$ for cubic lattices (first few). The diamond structure is fcc with the additional extinction of all-even hkl with $h + k + l = 4n + 2$ [2].

Lattice	Rule for hkl	Allowed N
Simple cubic (P)	none	1, 2, 3, 4, 5, 6, 8, 9, 10, 11, 12, …
Body-centred (I)	$h + k + l$ even	2, 4, 6, 8, 10, 12, 14, 16, …
Face-centred (F)	h, k, l all odd or all even	3, 4, 8, 11, 12, 16, 19, 20, 24, …
Diamond cubic	F, and not $h + k + l = 4n + 2$	3, 8, 11, 16, 19, 24, …

Once N_i is assigned, every peak gives a lattice parameter,

$$a_i = \frac{\lambda\sqrt{N_i}}{2\sin\theta_i}, \tag{5}$$

with $\lambda = 1.54056$ Å (Cu Kα_1). If σ_θ (rad) is the uncertainty of the Bragg angle, then

$$\sigma_{a_i} = \left|\frac{\partial a_i}{\partial\theta}\right|\sigma_\theta = a_i\cot\theta_i\,\sigma_\theta, \qquad \sigma_{\sin^2\theta} = \sin(2\theta)\,\sigma_\theta, \qquad \sigma_\theta = \tfrac{1}{2}\sigma_{2\theta}. \tag{6}$$

Kα doublet. Cu radiation contains Kα_1 ($\lambda_1 = 1.54056$ Å) and Kα_2 ($\lambda_2 = 1.54439$ Å) with intensity ratio $R \approx 0.5$. Differentiating Eq. (1) gives their angular separation,

$$\Delta(2\theta) \approx 2\tan\theta\,\frac{\lambda_2 - \lambda_1}{\lambda_1}, \tag{7}$$

which is 0.125° at $2\theta = 47.3^\circ$ and grows with angle, so the doublet is resolved only at high angle and only when the peaks are narrow.

Systematic angle errors. A zero-point error or a displaced specimen shifts all measured angles. Two standard ways of accounting for this are used:
(i) A global fit of all peaks (constant zero shift Δ_0) in which a and Δ_0 are the fit parameters:

$$2\theta_i^{\mathrm{obs}} = 2\arcsin\!\left(\frac{\lambda\sqrt{N_i}}{2a}\right) + \Delta_0, \tag{8}$$

(ii) The Nelson–Riley extrapolation [5], in which the a_i are fitted by a straight line

$$a_i = a_0 + m\,f(\theta_i), \qquad f(\theta) = \frac{1}{2}\left(\frac{\cos^2\theta}{\sin\theta} + \frac{\cos^2\theta}{\theta}\right), \tag{9}$$

and the intercept a_0 (the value at $f = 0$, i.e. $\theta = 90^\circ$) is taken as the corrected lattice parameter. Where the χ^2 of a fit exceeds its degrees of freedom ν, the parameter errors are multiplied by the Birge ratio [6] $\sqrt{\chi^2/\nu}$.

3 Experimental Setup and Procedure

3.1 Apparatus

The measurements were made on a powder X-ray diffractometer in Bragg–Brentano $\theta/2\theta$ geometry (Fig. 2), using Cu Kα radiation.

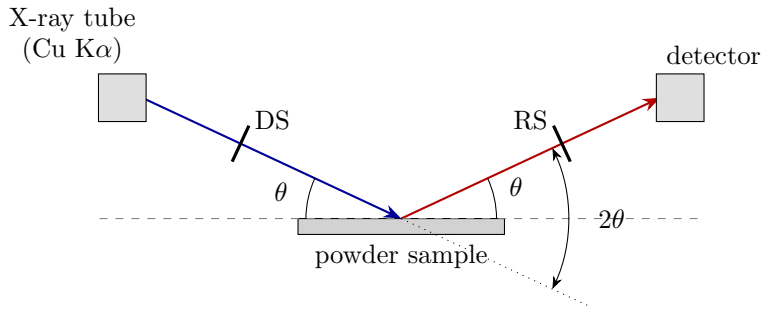


Figure 2: Schematic of the Bragg–Brentano $\theta/2\theta$ geometry. The X-ray tube and detector move through equal angles θ about the sample surface; the diffracted beam is deviated by 2θ from the incident direction. DS: divergence slit, RS: receiving slit.

3.2 Procedure

1. The silicon powder supplied for the experiment was ground to a fine, uniform powder, packed into the sample holder, and levelled flush with the holder surface.
2. The holder was mounted on the diffractometer, which was set up in $\theta/2\theta$ geometry with Cu Kα radiation.
3. The angle 2θ was scanned from 3° to 90° in steps of 0.01° and the intensity (cps) was recorded at each step.
4. Each pattern was exported as a two-column text file (2θ, intensity).
5. The peaks were located and fitted with Kα_1/Kα_2 pseudo-Voigt profiles (Sec. 5.1) the lattice parameter was obtained with the least-squares procedures of Sec. 5.2.

3.3 Sources of Uncertainty

Random (statistical): Counting noise limits the precision of each fitted peak position to $\sigma_{2\theta} = 0.0002$–$0.003°$. This was estimated from the covariance matrix of each fit, scaled by the residual variance. The step size of 0.01° is much smaller than the peak widths ($\Gamma \geq 0.09°$), so the angular sampling contributes no additional error beyond the fit.

Profile model: The Kα_2/Kα_1 ratio was fixed at $R = 0.5$. For the resolved Run 1 doublets between 47° and 88°, a fit with free R shifts the Kα_1 position by at most 0.001° and returns $R = 0.43$–0.50. For the unresolved (111) doublet of Run 1 and for all peaks of Run 2, R cannot be determined from the data and the fixed value was kept.

Systematic: Zero-point error, specimen displacement (a shift $\propto \cos\theta$), absorption/transparency and axial divergence (peak asymmetry at low angle) are the main instrumental effects. They were treated through the zero shift Δ_0 of Eq. (8) and the Nelson–Riley extrapolation, Eq. (9); the remaining systematic error is estimated from the scatter of the a_i and from the agreement with the literature value (Sec. 6). The sample temperature was not recorded; with $\alpha \approx 2.6 \times 10^{-6}$ K^{-1} for Si, a change of 1 K changes a by only 1.4×10^{-5} Å. The weak Kβ lines are below 0.5% of the Kα peaks and were ignored.

4 Observations and Data

Data files for the two runs contain 8701 points of 2θ (deg) and intensity (cps). The full patterns are shown in Fig. 3 and the raw peak data in Table 2.

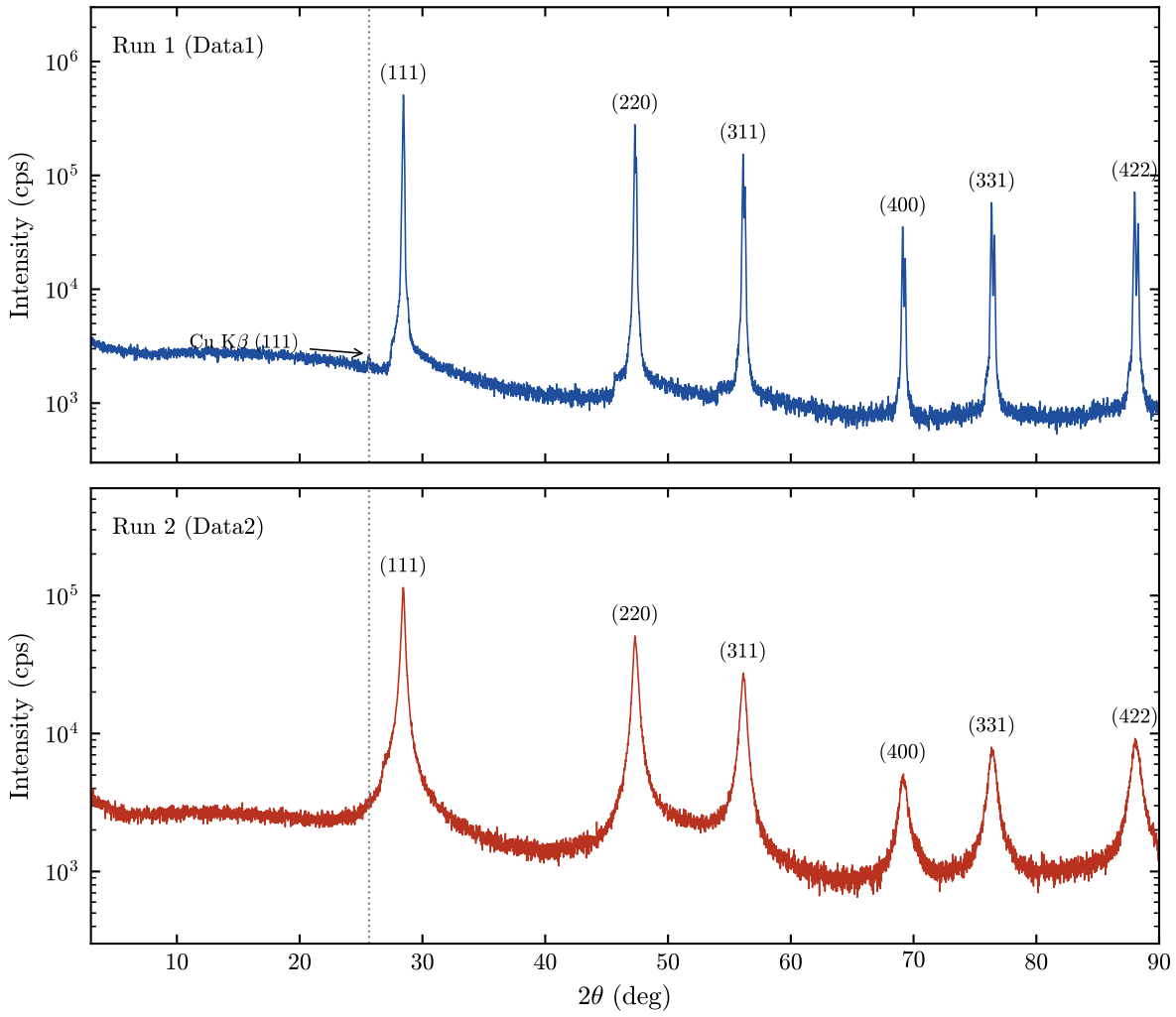


Figure 3: Measured powder patterns (logarithmic intensity scale) with the assigned Miller indices. The dotted line marks the expected position (25.65°) of the Cu Kβ line of the (111) reflection.

The following features were observed:

- Six Bragg peaks appear in each run, at $2\theta \approx 28.4°$, 47.3°, 56.1°, 69.1°, 76.4° and 88.0°; no peak is present below 28°.
- Run 1 has narrow peaks (FWHM 0.093–0.111°) with a maximum of 508830 cps at (111), and the Kα_1/Kα_2 doublets are visibly resolved from (220) upward. Run 2 has peaks 3 to 8 times wider (FWHM 0.305–0.789°), about 4.4 times lower in peak intensity (114400 cps), and unresolved doublets.
- Both patterns show a smoothly decreasing background from about 3.5×10^3 cps at 3° to about 10^3 cps at 60°–70°, and a weak bump at 25.6° at the position of the Cu Kβ line of (111) (calculated 25.65°).
- At the positions where (200), (222) and (420) would appear for $a = 5.431$ Å (32.96°, 58.86° and 78.74°), no intensity above the background is found.

Table 2: Position and height of the highest recorded channel of each Bragg peak (before profile fitting). Background was taken from the fitted linear background of Sec. 5.1. Intensities in counts per second (cps).

Peak	hkl	$2\theta_{\max}$ (°)	$I_{\max}$ (cps)	I_{bg} (cps)	$I_{\max} - I_{\mathrm{bg}}$ (cps)
Run 1 (Data1)					
1	(111)	28.45	508 830	2 668	506 162
2	(220)	47.30	278 705	1 599	277 106
3	(311)	56.11	153 592	1 291	152 300
4	(400)	69.11	35 222	797	34 424
5	(331)	76.36	57 680	851	56 829
6	(422)	88.01	71 353	912	70 442
Run 2 (Data2)					
1	(111)	28.44	114 400	5 295	109 105
2	(220)	47.31	51 000	2 563	48 437
3	(311)	56.14	27 433	1 775	25 658
4	(400)	69.17	5 100	928	4 172
5	(331)	76.36	7 967	996	6 971
6	(422)	88.04	9 217	1 203	8 014

5 Analysis and Results

5.1 Data Reduction

No smoothing, normalization or Kα_2 stripping was applied in the raw intensities. Each peak was isolated in a window around its maximum ($\pm 1.2°$ for Run 1; $\pm 1.8°$ to $\pm 3.1°$ for Run 2, see Table 10) and fitted by least squares with the Kα_1+Kα_2 model

$$I(2\theta) = b_0 + b_1(2\theta - x_0) + A\Big[p(2\theta; c_1, \Gamma, \eta) + R\,p\big(2\theta; c_2(c_1), \Gamma, \eta\big)\Big], \tag{10}$$

where b_0, b_1 describe a linear background, A is the Kα_1 amplitude, and $R = 0.5$. The profile $p = \eta L + (1-\eta)G$ is a normalized-height pseudo-Voigt function [7] (Lorentzian L, Gaussian G, common FWHM Γ, mixing parameter η). The Kα_2 position is tied to c_1 through Bragg's law,

$$c_2 = 2\arcsin\left[\frac{\lambda_2}{\lambda_1}\sin\frac{c_1}{2}\right], \tag{11}$$

so that only the Kα_1 position c_1 is free. The six free parameters per peak $(b_0, b_1, A, c_1, \Gamma, \eta)$ were found with SciPy's `curve_fit`; the quoted $\sigma_{2\theta}$ is the standard error of c_1 (covariance scaled by the residual variance, since the files give no count statistics). The results are in Table 3 and examples of the fits in Fig. 4. The fits reproduce the data with $R^2 \geq 0.98$; the residuals of Fig. 4 show small systematic misfits of the peak shape (about $\pm 10\%$ of the maximum for the strongest peak of Run 1), which are included in the error through the residual-variance scaling.

5.2 Fitting and Error Propagation

Indexing mathematically: The ratios $\sin^2\theta_i / \sin^2\theta_1$ are not integers, but multiplied by 3 they become integers within 0.007 (Run 1) and 0.043 (Run 2), giving $N = 3, 8, 11, 16, 19, 24$. SC lattice is excluded because it would require lines with $N = 1$ and 2 below 28°, which are absent; BCC is excluded because it allows only even N. The sequence 3, 8, 11, 16, 19, 24 is that of FCC (Table 1), so the peaks are (111), (220), (311), (400), (331) and (422).

Table 3: Fitted Kα_1 peak positions from the doublet pseudo-Voigt fits. $\sigma_{2\theta}$ is the standard error of c_1 from the covariance matrix (scaled by the residual variance); Γ is the common FWHM and η the Lorentzian fraction. Fit windows and all other parameters are listed in Table 10.

Peak	*hkl*	$2\theta_{K\alpha_1}$ (°)	$\sigma_{2\theta}$ (°)	Γ (°)	η	R^2
Run 1						
1	(111)	28.4310	0.0006	0.111±0.003	0.35	0.9872
2	(220)	47.2926	0.0004	0.093±0.001	0.45	0.9933
3	(311)	56.1089	0.0003	0.093±0.001	0.48	0.9956
4	(400)	69.1095	0.0003	0.098±0.001	0.54	0.9959
5	(331)	76.3536	0.0002	0.099±0.001	0.56	0.9965
6	(422)	88.0030	0.0002	0.102±0.001	0.64	0.9972
Run 2						
1	(111)	28.4109	0.0006	0.305±0.003	1.00	0.9971
2	(220)	47.2939	0.0007	0.425±0.003	1.00	0.9976
3	(311)	56.0874	0.0008	0.480±0.004	1.00	0.9969
4	(400)	69.1054	0.0026	0.764±0.012	1.00	0.9828
5	(331)	76.3475	0.0016	0.713±0.008	1.00	0.9926
6	(422)	88.0084	0.0019	0.789±0.010	1.00	0.9931

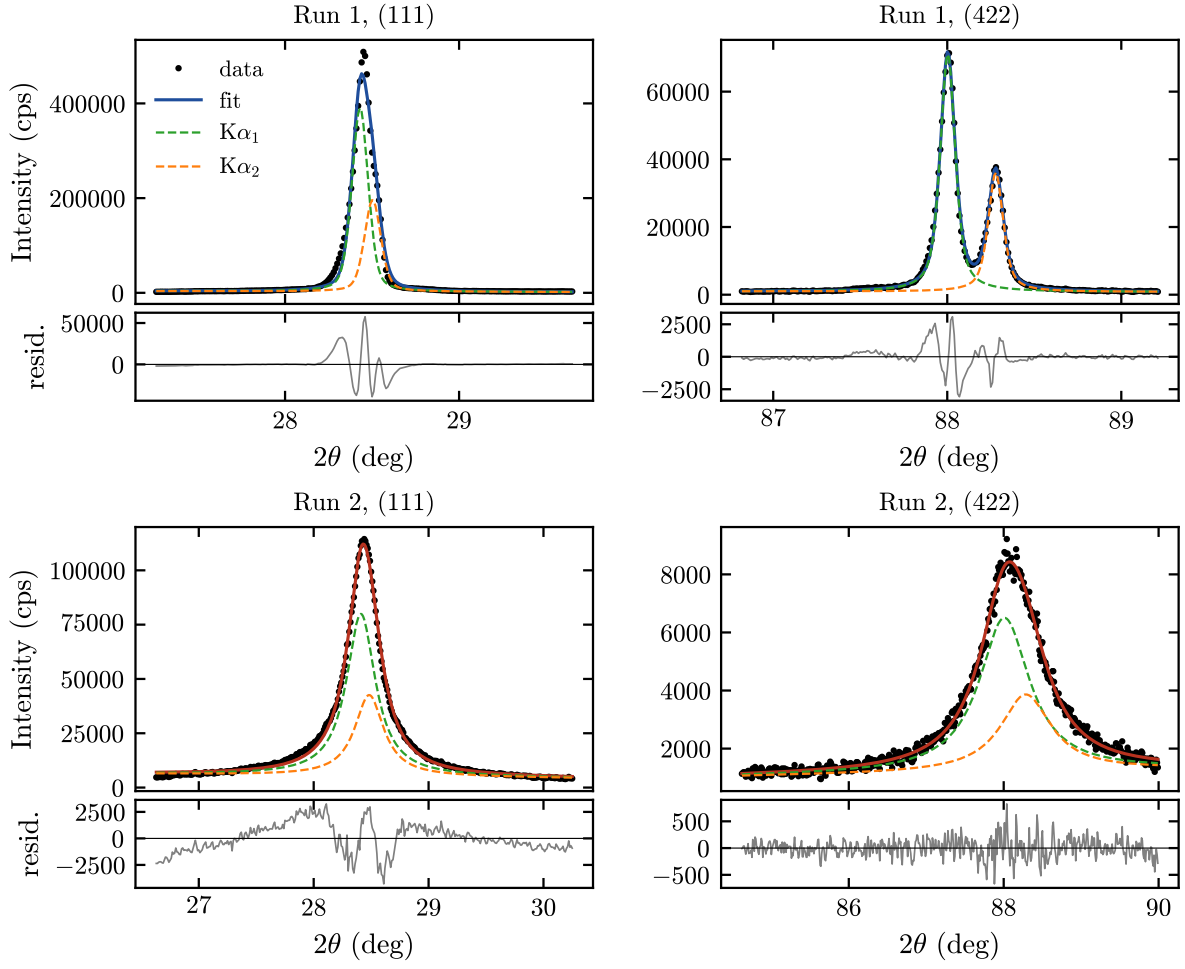


Figure 4: Doublet pseudo-Voigt fits for lowest-angle (111) and highest-angle (422) peaks of both runs, with Kα_1 and Kα_2 components, background and residuals. Kα doublet is resolved for (422) in Run 1 only.

Table 4: Mathematical indexing method, Run 1. The column 3× is the first multiple that gives integers (within ≤ 0.04), identifying $h^2+k^2+l^2$. $a = \lambda\sqrt{N}/(2\sin\theta)$ with $\lambda = 1.54056$ Å.

Peak	2θ (°)	$\sin^2\theta$	$1 \times \frac{\sin^2\theta}{\sin^2\theta_{\min}}$	2×	3×	$h^2+k^2+l^2$	hkl	a (Å)
1	28.4310	0.060305	1.0000	2.0000	3.0000	3	(111)	5.43293
2	47.2926	0.160873	2.6677	5.3353	8.0030	8	(220)	5.43191
3	56.1089	0.221192	3.6679	7.3358	11.0037	11	(311)	5.43201
4	69.1095	0.321708	5.3347	10.6694	16.0042	16	(400)	5.43222
5	76.3536	0.382035	6.3351	12.6702	19.0053	19	(331)	5.43217
6	88.0030	0.482577	8.0023	16.0046	24.0069	24	(422)	5.43214
						mean $\bar{a}\pm$ s.e.m.		5.43223 ± 0.00015

Table 5: Mathematical indexing method, Run 2.

Peak	2θ (°)	$\sin^2\theta$	$1 \times \frac{\sin^2\theta}{\sin^2\theta_{\min}}$	2×	3×	$h^2+k^2+l^2$	hkl	a (Å)
1	28.4109	0.060221	1.0000	2.0000	3.0000	3	(111)	5.43669
2	47.2939	0.160881	2.6715	5.3430	8.0145	8	(220)	5.43177
3	56.0874	0.221036	3.6704	7.3408	11.0112	11	(311)	5.43392
4	69.1054	0.321675	5.3416	10.6831	16.0247	16	(400)	5.43250
5	76.3475	0.381984	6.3430	12.6861	19.0291	19	(331)	5.43254
6	88.0084	0.482623	8.0142	16.0284	24.0426	24	(422)	5.43188
						mean $\bar{a}\pm$ s.e.m.		5.43322 ± 0.00076

Indexing analytically: The quotient search (Tables 13 and 14, Appendix A.4) shows that the lowest quotient shared by all peaks is $K_0 = \sin^2\theta_1/3$; the quotients $\sin^2\theta_1/1$ and $\sin^2\theta_1/2$ do not occur in the other columns. The peaks then correspond to $n = 3, 8, 11, 16, 19, 24$, the same integers as above. A weighted least-squares fit of $\sin^2\theta_i = KN_i$ gives $K = 0.0201073$ (Run 1) and 0.0201004 (Run 2).

Table 6: Analytical indexing method, Run 1. The last column is $a = \lambda\sqrt{N}/(2\sin\theta)$.

Peak	2θ (°)	$\sin^2\theta$	$\frac{\sin^2\theta}{2}$	$\frac{\sin^2\theta}{3}$	$\frac{\sin^2\theta}{K}$	$h^2+k^2+l^2$	hkl	a (Å)
1	28.4310	0.060305	0.030152	0.020102	2.9991	3	(111)	5.43293
2	47.2926	0.160873	0.080436	0.053624	8.0007	8	(220)	5.43191
3	56.1089	0.221192	0.110596	0.073731	11.0006	11	(311)	5.43201
4	69.1095	0.321708	0.160854	0.107236	15.9996	16	(400)	5.43222
5	76.3536	0.382035	0.191018	0.127345	18.9998	19	(331)	5.43217
6	88.0030	0.482577	0.241288	0.160859	24.0000	24	(422)	5.43214
						$a = \lambda/(2\sqrt{K})$		5.43215 ± 0.00004

Table 7: Analytical indexing method, Run 2. The last column is $a = \lambda\sqrt{N}/(2\sin\theta)$.

Peak	2θ (°)	$\sin^2\theta$	$\frac{\sin^2\theta}{2}$	$\frac{\sin^2\theta}{3}$	$\frac{\sin^2\theta}{K}$	$h^2+k^2+l^2$	hkl	a (Å)
1	28.4109	0.060221	0.030111	0.020074	2.9960	3	(111)	5.43669
2	47.2939	0.160881	0.080440	0.053627	8.0039	8	(220)	5.43177
3	56.0874	0.221036	0.110518	0.073679	10.9966	11	(311)	5.43392
4	69.1054	0.321675	0.160838	0.107225	16.0034	16	(400)	5.43250
5	76.3475	0.381984	0.190992	0.127328	19.0038	19	(331)	5.43254
6	88.0084	0.482623	0.241312	0.160874	24.0106	24	(422)	5.43188
						$a = \lambda/(2\sqrt{K})$		5.43308 ± 0.00070

Lattice parameter and error propagation: Complete calculation for every peak is in Appendix A. The statistical errors of the individual a_i (1×10^{-5} to 2×10^{-4} Å) are much smaller than the scatter between peaks (a_i spans 0.001 Å in Run 1 and 0.005 Å in Run 2). So a simple weighted mean would be misleading and the peak positions carry a systematic angular error. It was therefore treated with the global fit of Eq. (8) and the Nelson–Riley extrapolation of Eq. (9) (Figs. 5 and 6). The final values from the global fit, with the errors scaled by the Birge ratios 5.7 (Run 1) and 14.8 (Run 2), are

$$a_{\mathrm{Run\,1}} = 5.43220 \pm 0.00016 \text{ Å}, \qquad \Delta_0 = +0.001 \pm 0.002^\circ, \tag{12}$$

$$a_{\mathrm{Run\,2}} = 5.4304 \pm 0.0014 \text{ Å}, \qquad \Delta_0 = -0.028 \pm 0.014^\circ. \tag{13}$$

The two runs agree with each other: $a_{\mathrm{Run\,1}} - a_{\mathrm{Run\,2}} = 0.0018 \pm 0.0014$ Å, i.e. 1.3σ.

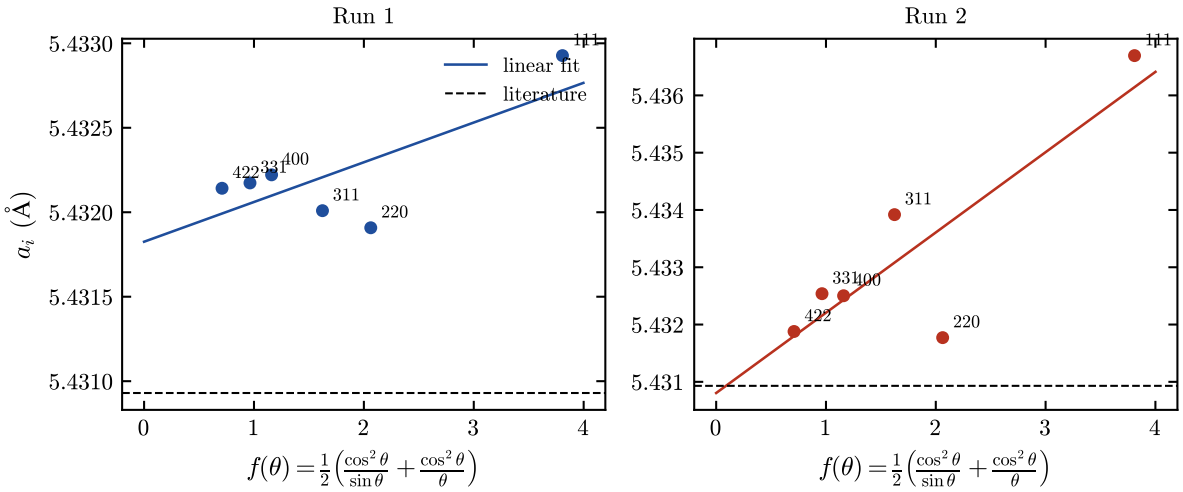


Figure 5: Per-peak lattice parameters a_i against the Nelson–Riley function $f(\theta)$ with the unweighted straight-line fit (solid) (literature value dashed). Error bars are $\leq 2 \times 10^{-4}$ Å).

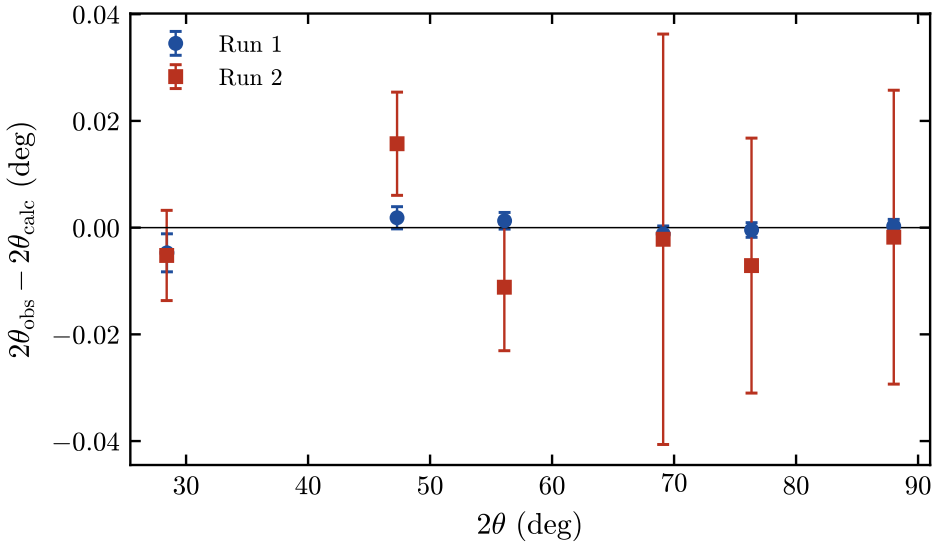


Figure 6: Residuals $2\theta_{\mathrm{obs}} - 2\theta_{\mathrm{calc}}$ of the global fit with zero shift. Error bars are 1σ fit errors scaled by the Birge ratio.

5.3 Result Summary

Table 8: Summary of the measured cubic lattice parameter a of Si compared with the literature value 5.43093 ± 0.00002 Å [4]. $\Delta a = a - a_{\mathrm{lit}}$; the last column is $\Delta a / \sqrt{\sigma_a^2 + \sigma_{\mathrm{lit}}^2}$.

Quantity: a (Å)	Measured value (Å)	Δa (10^{-3} Å)	% Deviation	σ-dev.
Run 1				
Mathematical method (mean of 6 peaks)	5.43223 ± 0.00015	+1.30	+0.024	+8.8
Analytical method ($\lambda/2\sqrt{K}$)	5.43215 ± 0.00004	+1.22	+0.022	+28.5
Nelson–Riley extrapolation a_0	5.43183 ± 0.00022	+0.90	+0.016	+4.1
Global fit with zero shift Δ_0	5.43220 ± 0.00016	+1.27	+0.023	+8.0
Run 2				
Mathematical method (mean of 6 peaks)	5.43322 ± 0.00076	+2.29	+0.042	+3.0
Analytical method ($\lambda/2\sqrt{K}$)	5.43308 ± 0.00070	+2.15	+0.040	+3.0
Nelson–Riley extrapolation a_0	5.43080 ± 0.00088	-0.13	-0.002	-0.1
Global fit with zero shift Δ_0	5.43040 ± 0.00139	-0.53	-0.010	-0.4

6 Discussion

- **Agreement with literature.** The Run 2 lattice parameter from the global fit (-0.010%) and the Nelson–Riley intercept (−0.002%) agree with the literature value [4] within 0.4σ and 0.1σ. The Run 1 value from the global fit lies 0.023% ($+1.27 \times 10^{-3}$ Å) above the literature value, i.e. 8.0σ in terms of its own statistical error, so Run 1 is accurate to 2×10^{-4} in relative terms but not within its quoted uncertainty. Mathematical and analytical methods give +0.02% (Run 1) and +0.04% (Run 2).
- **Dominant errors.** The errors are systematic rather than statistical. In Run 2 the a_i tend to decrease with increasing angle (Fig. 5), as expected from a zero shift or displacement of the specimen, and the fit finds $\Delta_0 = -0.028 \pm 0.014°$; once this is included the result agrees with the literature. In Run 1 five of the six a_i lie between 1.0 and 1.3×10^{-3} Å above a_{lit}, nearly independent of angle. This corresponds to a shift of 2θ proportional to $\tan\theta$, i.e. a uniform fractional scale error of about 2×10^{-4}. A zero shift or Nelson–Riley extrapolation cannot absorb such an error, which is why the Run 1 global fit has $\chi^2 = 128.6$ for $\nu = 4$. Its origin (effective wavelength, angle calibration, or a real difference in the lattice parameter of the sample) cannot be decided from these data alone, since no internal standard was measured.
- **Anomalies.** The (111) line deviates most in both runs ($a_1 - a_{\mathrm{lit}} = +2.0 \times 10^{-3}$ Å in Run 1, $+5.8 \times 10^{-3}$ Å in Run 2). It is the line most affected by axial divergence (asymmetry at low angle), by the unresolved Kα doublet, and, in Run 1, by its very high count rate (5×10^5 cps), which may saturate the detector. In Run 2 all fitted profiles have $\eta \to 1$, i.e. they are Lorentzian-like with heavy tails, and the FWHM grows from 0.31° to 0.79° with angle. This is characteristic of size or strain broadening, but it cannot be separated from the instrumental broadening easily.
- **Improvements.** (i) Measure an internal standard (e.g. a certified Si or LaB_6 powder) to calibrate the zero point and angular scale; (ii) use a flat, well-packed specimen at the correct height, preferably spinning; (iii) increase the counting time for the weak high-angle peaks and record the sample temperature; (iv) use a Kβ filter or monochromator; (v) extend the scan beyond 90° to the (511)/(333) and (440) lines at 94.9° and 106.7°, where the effect of angle errors on a_i is smallest and the Nelson–Riley extrapolation is better conditioned.

7 Conclusion

The powder diffraction patterns of silicon were indexed by the mathematical and the analytical method: the $\sin^2\theta$ ratios $3:8:11:16:19:24$ show FCC, and the absence of (200), (222) and (420) identifies the diamond structure. Fits of all six peaks with a zero shift give the lattice parameter $a = 5.43220 \pm 0.00016$ Å (Run 1) and 5.4304 ± 0.0014 Å (Run 2), compared with the literature value 5.43093 ± 0.00002 Å. Run 2 agrees within 0.4σ, while Run 1 is 0.023% above the literature value, which is outside its statistical error and indicates an uncorrected systematic error of about 1×10^{-3} Å. The aim of the experiment, to index the pattern and obtain the lattice parameter of a cubic crystal to better than 0.05%, was achieved.

References

[1] W. L. Bragg, "The diffraction of short electromagnetic waves by a crystal," *Proc. Camb. Phil. Soc.* **17**, 43 (1913).

[2] B. D. Cullity and S. R. Stock, *Elements of X-Ray Diffraction*, 3rd ed. (Prentice Hall, 2001).

[3] *X-ray diffraction (XRD) – Manual: Indexing a powder diffraction pattern*, student lab manual (course material).

[4] "Automated lattice parameter determination on single crystals," *Advances in X-ray Analysis* (International Centre for Diffraction Data, 1969): $a_0 = 5.43093 \pm 0.00002$ Å for silicon at 25°C.

[5] J. B. Nelson and D. P. Riley, "An experimental investigation of extrapolation methods in the derivation of accurate unit-cell dimensions of crystals," *Proc. Phys. Soc.* **57**, 160 (1945).

[6] R. T. Birge, "The calculation of errors by the method of least squares," *Phys. Rev.* **40**, 207 (1932).

[7] P. Thompson, D. E. Cox and J. B. Hastings, "Rietveld refinement of Debye–Scherrer synchrotron X-ray data from Al_2O_3," *J. Appl. Cryst.* **20**, 79 (1987).

A Raw Data and Worked Calculations

All numbers in this appendix come directly from the data files; $\lambda = 1.54056$ Å throughout.

A.1 Raw data and fit parameters

Experimental data has 8701 rows of 2θ (deg) and intensity (cps). Table 9 reproduces the raw channels around the (422) maximum, and Table 10 gives all fit parameters of Eq. (10).

Table 9: Raw intensities (cps) of both runs in the channels around the (422) peak ($2\theta = 87.94°$–$88.08°$).

2θ (°)	Run 1 (Data1)	Run 2 (Data2)
87.94	29831.7	8150.0
87.95	36061.7	7950.0
87.96	42886.7	8400.0
87.97	51205.0	8316.7
87.98	59756.7	8133.3
87.99	65835.0	7983.3
88.00	70676.7	8233.3
88.01	71353.3	8766.7
88.02	68553.3	8316.7
88.03	60981.7	8716.7
88.04	50750.0	9216.7
88.05	39421.7	8716.7
88.06	29971.7	8450.0
88.07	23345.0	8416.7

Table 10: Complete fit parameters of Eq. (10) for every peak window ($R = 0.5$ fixed): window range in 2θ (deg), number of channels n, background intercept b_0 and slope b_1 (cps and cps/deg), amplitude A (cps), Kα_1 position c_1 (deg), FWHM Γ (deg), Lorentzian fraction η, and rms residual (cps).

Peak	Window	n	b_0	b_1	A	c_1	Γ	η	rms
Run 1									
1	27.26–29.65	240	2668	-788	387462	28.4310	0.111	0.35	10742
2	46.11–48.49	239	1599	-373	265558	47.2926	0.093	0.45	4259
3	54.92–57.30	239	1291	-225	149196	56.1089	0.093	0.48	1896
4	67.91–70.31	241	797	-63	34130	69.1095	0.098	0.54	417
5	75.16–77.56	241	851	-69	55665	76.3536	0.099	0.56	625
6	86.82–89.21	240	912	-60	70319	88.0030	0.102	0.64	710
Run 2									
1	26.63–30.25	363	5295	-546	74671	28.4109	0.305	1.00	1347
2	44.99–49.63	465	2563	222	32657	47.2939	0.425	1.00	551
3	53.59–58.69	511	1775	-242	17787	56.0874	0.480	1.00	342
4	66.26–72.08	583	928	24	2645	69.1054	0.764	1.00	134
5	73.26–79.46	621	996	4	4580	76.3475	0.713	1.00	144
6	84.62–89.99	538	1203	47	5303	88.0084	0.789	1.00	174

A.2 Calculations of a and σ_a

The full chain from fitted 2θ to a is shown for the lowest- and highest-angle peak of each run; all peaks are tabulated in Sec. A.3.

Run 1, peak 1, (111)**.**

$$
\begin{aligned}
2\theta &= 28.43105^\circ \pm 0.00063^\circ \\
\theta &= \tfrac{1}{2}(2\theta) = 14.21552^\circ = 0.2481077 \text{ rad}, \qquad \sigma_\theta = \tfrac{1}{2}\sigma_{2\theta} = 5.483 \times 10^{-6} \text{ rad} \\
\sin\theta &= \sin(14.21552^\circ) = 0.2455700 \\
\sin^2\theta &= 0.2455700^2 = 0.0603046 \\
\sigma_{\sin^2\theta} &= \sin(2\theta)\,\sigma_\theta = 0.476101 \times 5.483 \times 10^{-6} = 2.611 \times 10^{-6} \\
\frac{\sin^2\theta}{\sin^2\theta_1} &= \frac{0.0603046}{0.0603046} = 1.00000, \qquad 3 \times 1.00000 = 3.0000 \;\Rightarrow\; N = 3 \\
d &= \frac{\lambda}{2\sin\theta} = \frac{1.54056}{2 \times 0.2455700} = 3.136702 \text{ Å} \\
a &= d\sqrt{N} = 3.136702 \times \sqrt{3} = 3.136702 \times 1.732051 = 5.43293 \text{ Å} \\
\sigma_a &= a\cot\theta\,\sigma_\theta = 5.43293 \times 3.94746 \times 5.483 \times 10^{-6} = 1.18 \times 10^{-4} \text{ Å}
\end{aligned}
$$

Run 1, peak 6, (422)**.**

$$
\begin{aligned}
2\theta &= 88.00302^\circ \pm 0.00022^\circ \\
\theta &= \tfrac{1}{2}(2\theta) = 44.00151^\circ = 0.7679712 \text{ rad}, \qquad \sigma_\theta = \tfrac{1}{2}\sigma_{2\theta} = 1.938 \times 10^{-6} \text{ rad} \\
\sin\theta &= \sin(44.00151^\circ) = 0.6946773 \\
\sin^2\theta &= 0.6946773^2 = 0.4825766 \\
\sigma_{\sin^2\theta} &= \sin(2\theta)\,\sigma_\theta = 0.999393 \times 1.938 \times 10^{-6} = 1.936 \times 10^{-6} \\
\frac{\sin^2\theta}{\sin^2\theta_1} &= \frac{0.4825766}{0.0603046} = 8.00231, \qquad 3 \times 8.00231 = 24.0069 \;\Rightarrow\; N = 24 \\
d &= \frac{\lambda}{2\sin\theta} = \frac{1.54056}{2 \times 0.6946773} = 1.108831 \text{ Å} \\
a &= d\sqrt{N} = 1.108831 \times \sqrt{24} = 1.108831 \times 4.898979 = 5.43214 \text{ Å} \\
\sigma_a &= a\cot\theta\,\sigma_\theta = 5.43214 \times 1.03548 \times 1.938 \times 10^{-6} = 1.09 \times 10^{-5} \text{ Å}
\end{aligned}
$$

Run 2, peak 1, (111)**.**

$$
\begin{aligned}
2\theta &= 28.41093^\circ \pm 0.00057^\circ \\
\theta &= \tfrac{1}{2}(2\theta) = 14.20547^\circ = 0.2479322 \text{ rad}, \qquad \sigma_\theta = \tfrac{1}{2}\sigma_{2\theta} = 4.979 \times 10^{-6} \text{ rad} \\
\sin\theta &= \sin(14.20547^\circ) = 0.2453999 \\
\sin^2\theta &= 0.2453999^2 = 0.0602211 \\
\sigma_{\sin^2\theta} &= \sin(2\theta)\,\sigma_\theta = 0.475792 \times 4.979 \times 10^{-6} = 2.369 \times 10^{-6} \\
\frac{\sin^2\theta}{\sin^2\theta_1} &= \frac{0.0602211}{0.0602211} = 1.00000, \qquad 3 \times 1.00000 = 3.0000 \;\Rightarrow\; N = 3 \\
d &= \frac{\lambda}{2\sin\theta} = \frac{1.54056}{2 \times 0.2453999} = 3.138877 \text{ Å} \\
a &= d\sqrt{N} = 3.138877 \times \sqrt{3} = 3.138877 \times 1.732051 = 5.43669 \text{ Å} \\
\sigma_a &= a\cot\theta\,\sigma_\theta = 5.43669 \times 3.95038 \times 4.979 \times 10^{-6} = 1.07 \times 10^{-4} \text{ Å}
\end{aligned}
$$

Run 2, peak 6, (422)**.**

$$
\begin{aligned}
2\theta &= 88.00838^\circ \pm 0.00186^\circ \\
\theta &= \tfrac{1}{2}(2\theta) = 44.00419^\circ = 0.7680180 \text{ rad}, \qquad \sigma_\theta = \tfrac{1}{2}\sigma_{2\theta} = 1.622 \times 10^{-5} \text{ rad} \\
\sin\theta &= \sin(44.00419^\circ) = 0.6947110 \\
\sin^2\theta &= 0.6947110^2 = 0.4826233 \\
\sigma_{\sin^2\theta} &= \sin(2\theta)\,\sigma_\theta = 0.999396 \times 1.622 \times 10^{-5} = 1.621 \times 10^{-5} \\
\frac{\sin^2\theta}{\sin^2\theta_1} &= \frac{0.4826233}{0.0602211} = 8.01419, \qquad 3 \times 8.01419 = 24.0426 \;\Rightarrow\; N = 24 \\
d &= \frac{\lambda}{2\sin\theta} = \frac{1.54056}{2 \times 0.6947110} = 1.108778 \text{ Å} \\
a &= d\sqrt{N} = 1.108778 \times \sqrt{24} = 1.108778 \times 4.898979 = 5.43188 \text{ Å} \\
\sigma_a &= a\cot\theta\,\sigma_\theta = 5.43188 \times 1.03538 \times 1.622 \times 10^{-5} = 9.12 \times 10^{-5} \text{ Å}
\end{aligned}
$$

A.3 Per-peak results

Table 11: Complete per-peak calculation, Run 1: $\theta = (2\theta)/2$, $\sin\theta$, $\sin^2\theta$, $d = \lambda/(2\sin\theta)$, $N = h^2+k^2+l^2$, $a = d\sqrt{N}$ and $\sigma_a = a\cot\theta\,\sigma_\theta$ (in units of 10^{-5} Å).

Peak	2θ (°)	$\sigma_{2\theta}$ (°)	θ (°)	$\sin\theta$	$\sin^2\theta$	d (Å)	N	a (Å)	σ_a
1	28.4310	0.0006	14.21552	0.245570	0.060305	3.13670	3	5.43293	11.8
2	47.2926	0.0004	23.64630	0.401089	0.160873	1.92047	8	5.43191	4.0
3	56.1089	0.0003	28.05444	0.470310	0.221192	1.63781	11	5.43201	2.4
4	69.1095	0.0003	34.55474	0.567193	0.321708	1.35806	16	5.43222	1.8
5	76.3536	0.0002	38.17680	0.618090	0.382035	1.24623	19	5.43217	1.5
6	88.0030	0.0002	44.00151	0.694677	0.482577	1.10883	24	5.43214	1.1

Table 12: Complete per-peak calculation, Run 2

Peak	2θ (°)	$\sigma_{2\theta}$ (°)	θ (°)	$\sin\theta$	$\sin^2\theta$	d (Å)	N	a (Å)	σ_a
1	28.4109	0.0006	14.20547	0.245400	0.060221	3.13888	3	5.43669	10.7
2	47.2939	0.0007	23.64693	0.401099	0.160881	1.92042	8	5.43177	7.1
3	56.0874	0.0008	28.04371	0.470145	0.221036	1.63839	11	5.43392	7.1
4	69.1054	0.0026	34.55270	0.567164	0.321675	1.35813	16	5.43250	17.9
5	76.3475	0.0016	38.17377	0.618049	0.381984	1.24631	19	5.43254	9.7
6	88.0084	0.0019	44.00419	0.694711	0.482623	1.10878	24	5.43188	9.1

A.4 Quotient search (analytical method)

Table 13: Quotient search Run 1: $\sin^2\theta_i/n$ for $n = 1, \ldots, 24$. Bold entries lie within 1% of the common quotient $K_0 = \sin^2\theta_1/3 = 0.020102$; K_0 is the lowest value that appears in every column, once each.

n	Peak 1	Peak 2	Peak 3	Peak 4	Peak 5	Peak 6
1	0.06030	0.16087	0.22119	0.32171	0.38204	0.48258
2	0.03015	0.08044	0.11060	0.16085	0.19102	0.24129
3	**0.02010**	0.05362	0.07373	0.10724	0.12735	0.16086
4	0.01508	0.04022	0.05530	0.08043	0.09551	0.12064
5	0.01206	0.03217	0.04424	0.06434	0.07641	0.09652
6	0.01005	0.02681	0.03687	0.05362	0.06367	0.08043
7	0.00861	0.02298	0.03160	0.04596	0.05458	0.06894
8	0.00754	**0.02011**	0.02765	0.04021	0.04775	0.06032
9	0.00670	0.01787	0.02458	0.03575	0.04245	0.05362
10	0.00603	0.01609	0.02212	0.03217	0.03820	0.04826
11	0.00548	0.01462	**0.02011**	0.02925	0.03473	0.04387
12	0.00503	0.01341	0.01843	0.02681	0.03184	0.04021
13	0.00464	0.01237	0.01701	0.02475	0.02939	0.03712
14	0.00431	0.01149	0.01580	0.02298	0.02729	0.03447
15	0.00402	0.01072	0.01475	0.02145	0.02547	0.03217
16	0.00377	0.01005	0.01382	**0.02011**	0.02388	0.03016
17	0.00355	0.00946	0.01301	0.01892	0.02247	0.02839
18	0.00335	0.00894	0.01229	0.01787	0.02122	0.02681
19	0.00317	0.00847	0.01164	0.01693	**0.02011**	0.02540
20	0.00302	0.00804	0.01106	0.01609	0.01910	0.02413
21	0.00287	0.00766	0.01053	0.01532	0.01819	0.02298
22	0.00274	0.00731	0.01005	0.01462	0.01737	0.02194
23	0.00262	0.00699	0.00962	0.01399	0.01661	0.02098
24	0.00251	0.00670	0.00922	0.01340	0.01592	**0.02011**

Table 14: Quotient search for analytical method, Run 2.

n	Peak 1	Peak 2	Peak 3	Peak 4	Peak 5	Peak 6
1	0.06022	0.16088	0.22104	0.32168	0.38198	0.48262
2	0.03011	0.08044	0.11052	0.16084	0.19099	0.24131
3	**0.02007**	0.05363	0.07368	0.10723	0.12733	0.16087
4	0.01506	0.04022	0.05526	0.08042	0.09550	0.12066
5	0.01204	0.03218	0.04421	0.06434	0.07640	0.09652
6	0.01004	0.02681	0.03684	0.05361	0.06366	0.08044
7	0.00860	0.02298	0.03158	0.04595	0.05457	0.06895
8	0.00753	**0.02011**	0.02763	0.04021	0.04775	0.06033
9	0.00669	0.01788	0.02456	0.03574	0.04244	0.05362
10	0.00602	0.01609	0.02210	0.03217	0.03820	0.04826
11	0.00547	0.01463	**0.02009**	0.02924	0.03473	0.04387
12	0.00502	0.01341	0.01842	0.02681	0.03183	0.04022
13	0.00463	0.01238	0.01700	0.02474	0.02938	0.03712
14	0.00430	0.01149	0.01579	0.02298	0.02728	0.03447
15	0.00401	0.01073	0.01474	0.02145	0.02547	0.03217
16	0.00376	0.01006	0.01381	**0.02010**	0.02387	0.03016
17	0.00354	0.00946	0.01300	0.01892	0.02247	0.02839
18	0.00335	0.00894	0.01228	0.01787	0.02122	0.02681
19	0.00317	0.00847	0.01163	0.01693	**0.02010**	0.02540
20	0.00301	0.00804	0.01105	0.01608	0.01910	0.02413
21	0.00287	0.00766	0.01053	0.01532	0.01819	0.02298
22	0.00274	0.00731	0.01005	0.01462	0.01736	0.02194
23	0.00262	0.00699	0.00961	0.01399	0.01661	0.02098
24	0.00251	0.00670	0.00921	0.01340	0.01592	**0.02011**

A.5 Least-squares calculations

RUN 1

(a) Common quotient K (analytical method). With weights $w_i = 1/\sigma^2_{\sin^2\theta_i}$ (Tab. 11),

$$K = \frac{\sum w_i \sin^2\theta_i\, N_i}{\sum w_i N_i^2} = \frac{6.792\times10^{12}}{3.378\times10^{14}} = 0.02010733,$$

$$\sigma_K^{\mathrm{stat}} = \Big(\sum w_i N_i^2\Big)^{-1/2} = 5.44\times10^{-8}$$

$$\chi^2 = \sum w_i(\sin^2\theta_i - K N_i)^2 = 132.0\ (\nu=5), \qquad \sqrt{\chi^2/\nu} = 5.14$$

$$a = \frac{\lambda}{2\sqrt{K}} = \frac{1.54056}{2\times0.1418003} = 5.43215\ \text{Å},$$

$$\sigma_a = \frac{a\,\sigma_K}{2K}\sqrt{\chi^2/\nu} = \frac{5.43215\times5.44\times10^{-8}}{2\times0.020107}\times5.14 = 3.8\times10^{-5}\ \text{Å}$$

(b) Mean of the per-peak values (mathematical method).

$$\bar a = \frac{1}{n}\sum a_i = 5.43223\ \text{Å}, \qquad s = \sqrt{\frac{\sum(a_i-\bar a)^2}{n-1}} = 0.00036\ \text{Å}, \qquad \sigma_{\bar a} = \frac{s}{\sqrt{n}} = 0.00015\ \text{Å}.$$

(c) Nelson–Riley extrapolation. $a_i = a_0 + m\,f(\theta_i)$ with $u_i = (a_i - 5.4300\,\text{Å})\times10^3$ mÅ:

i	f_i	a_i (Å)	u_i	f_i^2	$u_i f_i$	$u_i - \hat u_i$
1	3.80702	5.43293	2.927	14.49339	11.1439	+0.206
2	2.06268	5.43191	1.908	4.25464	3.9359	-0.402
3	1.62326	5.43201	2.009	2.63496	3.2616	-0.198
4	1.16028	5.43222	2.222	1.34625	2.5776	+0.123
5	0.96362	5.43217	2.174	0.92856	2.0944	+0.121
6	0.70930	5.43214	2.142	0.50310	1.5193	+0.150
Σ	10.32615		13.382	24.16090	24.5327	

$$D = n\sum f_i^2 - \Big(\sum f_i\Big)^2 = 6\times24.16090 - 10.32615^2 = 38.33608$$

$$m = \frac{n\sum u_i f_i - \sum f_i \sum u_i}{D} = \frac{6\times24.5327 - 10.32615\times13.382}{38.33608} = 0.2352\ \text{mÅ}$$

$$a_0 = 5.4300 + \frac{\sum u_i - m'\sum f_i}{n}\times10^{-3} = 5.43183\ \text{Å} \quad (m' = 0.2352)$$

$$s_r^2 = \frac{\sum(a_i - a_0 - m f_i)^2}{n-2} = (0.2720\ \text{mÅ})^2$$

$$\sigma_{a_0} = s_r\sqrt{\frac{\sum f_i^2}{D}} = 0.2160\ \text{mÅ}, \qquad \sigma_m = s_r\sqrt{\frac{n}{D}} = 0.1076\ \text{mÅ}$$

(d) Global fit with zero shift. Model $2\theta_i^{\mathrm{calc}} = 2\arcsin\!\big(\lambda\sqrt{N_i}/2a\big) + \Delta_0$, minimizing $\chi^2 = \sum[(2\theta_i - 2\theta_i^{\mathrm{calc}})/\sigma_i]^2$ (Gauss–Newton, converged to $< 10^{-12}$). Jacobian columns: $\partial(2\theta^{\mathrm{calc}})/\partial a = -(2\tan\theta_i/a)(180/\pi)$ deg/Å, $\partial/\partial\Delta_0 = 1$. With J_{ij} divided by σ_i:

$$J^TJ = \begin{pmatrix} 1.851\times10^{10} & -1.135\times10^{9} \\ -1.135\times10^{9} & 7.482\times10^{7}\end{pmatrix}$$

$$\mathrm{Cov} = (J^TJ)^{-1} \;\Rightarrow\; \sigma_a^{\mathrm{stat}} = 2.78\times10^{-5}\ \text{Å}, \quad \sigma_{\Delta_0}^{\mathrm{stat}} = \big(4.36\times10^{-4}\big)^\circ$$

$$\chi^2 = 128.6\ (\nu=4) \;\Rightarrow\; \sqrt{\chi^2/\nu} = 5.67$$

$$a = 5.43220 \pm 2.78\times10^{-5}\times5.67 = 5.43220 \pm 0.00016\ \text{Å}$$

$$\Delta_0 = +0.0008^\circ \pm 0.0025^\circ$$

(e) Deviation from literature.

$$\frac{a - a_{\mathrm{lit}}}{\sqrt{\sigma_a^2 + \sigma_{\mathrm{lit}}^2}} = \frac{5.43220 - 5.43093}{\sqrt{0.00016^2 + 0.00002^2}} = +7.98, \qquad \frac{\Delta a}{a_{\mathrm{lit}}} = +0.0233\%.$$

RUN 2

(a) Common quotient K (analytical method). With weights $w_i = 1/\sigma^2_{\sin^2\theta_i}$ (Tab. 12),

$$K = \frac{\sum w_i \sin^2\theta_i\, N_i}{\sum w_i N_i^2} = \frac{2.720\times10^{11}}{1.353\times10^{13}} = 0.02010042,$$

$$\sigma_K^{\mathrm{stat}} = \Big(\sum w_i N_i^2\Big)^{-1/2} = 2.72\times10^{-7}$$

$$\chi^2 = \sum w_i(\sin^2\theta_i - K N_i)^2 = 1840.0\ (\nu=5), \qquad \sqrt{\chi^2/\nu} = 19.18$$

$$a = \frac{\lambda}{2\sqrt{K}} = \frac{1.54056}{2\times0.1417759} = 5.43308\ \text{Å},$$

$$\sigma_a = \frac{a\,\sigma_K}{2K}\sqrt{\chi^2/\nu} = \frac{5.43308\times2.72\times10^{-7}}{2\times0.020100}\times19.18 = 7.0\times10^{-4}\ \text{Å}$$

(b) Mean of the per-peak values (mathematical method).

$$\bar a = \frac{1}{n}\sum a_i = 5.43322\ \text{Å}, \qquad s = \sqrt{\frac{\sum(a_i-\bar a)^2}{n-1}} = 0.00187\ \text{Å}, \qquad \sigma_{\bar a} = \frac{s}{\sqrt{n}} = 0.00076\ \text{Å}.$$

(c) Nelson–Riley extrapolation. $a_i = a_0 + m\,f(\theta_i)$ with $u_i = (a_i - 5.4300\,\text{Å})\times10^3$ mÅ

i	f_i	a_i (Å)	u_i	f_i^2	$u_i f_i$	$u_i - \hat u_i$
1	3.81002	5.43669	6.694	14.51629	25.5055	+0.550
2	2.06260	5.43177	1.772	4.25433	3.6540	-1.924
3	1.62417	5.43392	3.919	2.63794	6.3644	+0.838
4	1.16040	5.43250	2.503	1.34653	2.9042	+0.072
5	0.96377	5.43254	2.539	0.92885	2.4468	+0.383
6	0.70919	5.43188	1.879	0.50296	1.3325	+0.080
Σ	10.33017		19.305	24.18691	42.2074	

$$D = n\sum f_i^2 - \Big(\sum f_i\Big)^2 = 6\times24.18691 - 10.33017^2 = 38.40906$$

$$m = \frac{n\sum u_i f_i - \sum f_i \sum u_i}{D} = \frac{6\times42.2074 - 10.33017\times19.305}{38.40906} = 1.4013\ \text{mÅ}$$

$$a_0 = 5.4300 + \frac{\sum u_i - m'\sum f_i}{n}\times10^{-3} = 5.43080\ \text{Å} \quad (m' = 1.4013)$$

$$s_r^2 = \frac{\sum(a_i - a_0 - m f_i)^2}{n-2} = (1.1027\ \text{mÅ})^2$$

$$\sigma_{a_0} = s_r\sqrt{\frac{\sum f_i^2}{D}} = 0.8750\ \text{mÅ}, \qquad \sigma_m = s_r\sqrt{\frac{n}{D}} = 0.4358\ \text{mÅ}$$

(d) Global fit with zero shift. Model $2\theta_i^{\mathrm{calc}} = 2\arcsin\!\big(\lambda\sqrt{N_i}/2a\big) + \Delta_0$, minimizing $\chi^2 = \sum[(2\theta_i - 2\theta_i^{\mathrm{calc}})/\sigma_i]^2$ (Gauss–Newton, converged to $< 10^{-12}$). Jacobian columns: $\partial(2\theta^{\mathrm{calc}})/\partial a = -(2\tan\theta_i/a)(180/\pi)$ deg/Å, $\partial/\partial\Delta_0 = 1$. With J_{ij} divided by σ_i:

$$J^TJ = \begin{pmatrix} 7.426\times10^{8} & -7.003\times10^{7} \\ -7.003\times10^{7} & 7.794\times10^{6}\end{pmatrix}$$

$$\mathrm{Cov} = (J^TJ)^{-1} \ \Rightarrow\ \sigma_a^{\mathrm{stat}} = 9.39\times10^{-5}\ \text{Å}, \quad \sigma_{\Delta_0}^{\mathrm{stat}} = \big(9.17\times10^{-4}\big)^\circ$$

$$\chi^2 = 878.5\ (\nu=4)\ \Rightarrow\ \sqrt{\chi^2/\nu} = 14.82$$

$$a = 5.43040 \pm 9.39\times10^{-5}\times14.82 = 5.43040 \pm 0.00139\ \text{Å}$$

$$\Delta_0 = -0.0284^\circ \pm 0.0136^\circ$$

(e) Deviation from literature.

$$\frac{a - a_{\mathrm{lit}}}{\sqrt{\sigma_a^2 + \sigma_{\mathrm{lit}}^2}} = \frac{5.43040 - 5.43093}{\sqrt{0.00139^2 + 0.00002^2}} = -0.38, \qquad \frac{\Delta a}{a_{\mathrm{lit}}} = -0.0098\%.$$